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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Student Academic Performance Management System with Grade Calculation

Submitted by

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Under the Guidance of

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: II

DECLARATION

I hereby declare that the project entitled "**student Academic Performance Management system with grade calculation**" submitted in partial fulfillment of the requirements for the course BSc Information Technology is a **record of my original work** carried out under the guidance of **Mugilan R** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Arthi B

CERTIFICATE

This is to certify that the project entitled **"Student Academic Performance Management System with Grade Calculation"** is a **bona fide work** carried out by Arthi B, B.Sc.Information Technology - 'B' ,**KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva -Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Student Academic Performance Management System with Grade Calculation is a software system designed to manage and analyze student academic performance efficiently. The main purpose of this system is to store, organize, and evaluate students' academic records in a structured way.

In traditional systems, managing student marks and calculating grades manually can take more time and may lead to errors. This system helps teachers or administrators to easily enter student details, record subject marks, and automatically calculate the total marks, percentage, and grade of each student.

The system provides a simple and user-friendly interface to maintain student information such as name, register number, subjects, and marks. Based on the marks entered, the system calculates the grade using predefined grading rules. It also helps in generating reports that show the academic performance of students.

This project improves accuracy, saves time, and makes academic record management easier. Overall, the Student Academic Performance Management System helps educational institutions manage student performance effectively and efficiently.

1.INTRODUCTION

In educational institutions, managing student academic records is an important task. Schools and colleges need an efficient system to store, organize, and evaluate student performance. In many cases, student marks and grades are still managed manually using registers or spreadsheets. This method can be time-consuming, difficult to maintain, and may lead to calculation errors.

The Student Academic Performance Management System with Grade Calculation is designed to simplify the process of managing student academic information. This system allows teachers or administrators to enter student details, subject marks, and automatically calculate the total marks, percentage, and grade.

The main aim of this system is to provide a simple and reliable way to monitor the academic performance of students. It helps in reducing manual work, improving accuracy, and saving time. The system also helps in generating student performance reports quickly and efficiently.

By using this system, educational institutions can maintain student academic records in a structured and organized way. This makes it easier to track student progress and analyze their performance over time.

1.1 **Objective**.

The main objective of the **Student Academic Performance Management System with Grade Calculation** is to develop a simple and efficient application using Core Java that helps in managing student academic records and automatically calculating grades based on their marks.

The specific objectives of this project are:

- To design and develop a **Java-based student performance management system.**

- To **store and manage student academic details** such as name, register number, subjects, and marks.
- To **automate the calculation of total marks, percentage, and grade** of each student.
- To **reduce manual errors** in mark calculation and grade evaluation.
- To provide a **user-friendly system for teachers or administrators** to manage student records easily. To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements.

1.2 Overview

The Student Academic Performance Management System with Grade Calculation is designed to help educational institutions manage student academic information in a simple and organized way. This system allows teachers or administrators to store student details, enter subject marks, and automatically calculate the total marks, percentage, and grade.

In many institutions, student performance is still managed manually, which may lead to errors and difficulty in maintaining records. This system helps to overcome those problems by providing a computerized method to manage student academic data.

The system uses Core Java programming concepts to develop a simple and efficient application. It allows users to add student details, update marks, and view the academic performance of students. The system also calculates grades based on the marks entered.

Overall, this project helps in reducing manual work, improving accuracy, and providing a clear view of student academic performance. It makes the management of student records easier and more efficient for educational institutions.

1.3 Features

The main features of the **Student Academic Performance Management System with Grade Calculation** are:

- **Student Information Management** – The system allows storing and managing student details such as name, register number, and course information.
- **Marks Entry** – Teachers or administrators can easily enter and update marks for different subjects.
- **Automatic Grade Calculation** – The system automatically calculates the grade based on the marks entered.
- **Total and Percentage Calculation** – It calculates the total marks and percentage of each student automatically.
- **Data Accuracy** – The system reduces manual calculation errors and improves accuracy.
- **User-Friendly Interface** – The application is simple and easy to use for teachers and administrators.

PROGRAM ALALYSIS

Performance Report Generation – The system can generate reports that show the academic performance of students.

Program analysis helps to understand the requirements and working process of the Student Academic

Performance Management System with Grade Calculation. In this system, student details and subject marks are entered and stored in the program.

The system automatically calculates total marks, percentage, and grade using Core Java concepts like classes, methods, loops, and conditions. This reduces manual work and improves accuracy.

2.1 Problem Definition

In many schools and colleges, student academic performance is often managed manually using registers, files, or basic spreadsheets. This traditional method creates several challenges.

Some common problems include:

- Errors in calculating total marks and grade.
- Time-consuming record management
- Difficulty in maintaining student data
- Lack of proper performance reports
- Manual systems increase the workload for teachers and may lead to mistakes in grade calculation.

Therefore, there is a need for a computerized system to manage student academic records and calculate grades automatically. The Student Academic Performance Management System with Grade Calculation helps to solve these problems by providing an easy and efficient solution.

2.2 Project Overview

The Student Academic Performance Management System with Grade Calculation is a Java-based application developed to manage student academic records easily and efficiently.

The system allows users to enter student details and subject marks. Based on the marks entered, the system automatically calculates the total marks, percentage, and grade of each student.

This system helps to reduce manual work, improve accuracy, and maintain student performance records in an organized manner.

This project is developed using Core Java concepts such as classes, objects, methods, loops, and conditional statements. It provides a simple and user-friendly system for teachers or administrators to manage and analyze student academic performance effectively.

2.3 Scopes and Limitations

Scope

The scope of the Student Academic Performance Management System with Grade Calculation includes the following:

- To store and manage student details such as name, register number, and subject information.
- To enter and maintain subject marks for each student.To automatically calculate total marks, percentage, and grade based on the marks entered.
- To reduce manual calculation errors and save time in result preparation.
- To help teachers or administrators easily monitor and analyze student academic performance.

Limitations.

The limitations of the Student Academic Performance Management System with Grade Calculation include the following:

- The system is designed only for basic student record and grade calculation.
- It does not support online access for students or teachers.
- The system may handle only a limited number of student records.
- It does not include advanced report generation features.
- The system depends on manual entry of student marks and details.

3.SYSTEM DESIGN

The main modules of the **Student Academic Performance Management System with Grade Calculation** are explained below:

1. Student Information Module

This module is responsible for storing basic student details such as name, register number, and course information.

Functions of this module include:

- Accepting student details
- Storing student information
- Managing student records

Example Student Details:

Name: Arthi

Register Number: 12345

Course: B.Sc. IT

2. Marks Entry Module

This module allows the user to enter subject marks for each student.

Functions of this module include:

- Accepting subject names
- Entering marks for each subject
- Storing the marks for calculation

Example Subjects:

1.Tamil

2.English

3.Mathematics

4.ComputerScience

3. Grade Calculation module

This module performs the calculation of total marks, percentage, and grade based on the marks entered.

The system calculates:

- Total marks
- Percentage
- Grade of the student

Example Calculation:

Total Marks = Sum of all subject marks

Percentage = (Total Marks / Total Subjects)

Grade = Based on percentage

4. Result Display Module

This module displays the academic performance of the student.

The result includes:

- Student name

- Subject marks
- Total marks
- Percentage
- Grade

Example Output:

------ STUDENT RESULT -----

Name: Arthi B

Total Marks: 450

Percentage: 90%

Grade: A

5. Exit Module

This module allows the user to safely exit the application after completing the process.

When the user selects the exit option, the system closes the program.

1.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Student Academic Performance Management System with Grade Calculation, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of managing student academic records.

The main problems identified in manual academic record systems are:

- Manual calculation may lead to errors in marks and grades.
- Managing student records manually takes more time.
- Difficulty in maintaining and organizing academic data.
- Lack of proper performance reports for students.

Based on these problems, the system requirements were defined to develop an automated system that simplifies the management of student academic performance.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Student Information Module

- Marks Entry Module
- Grade Calculation Module
- Result Display Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of logic for storing student details
- Writing code for marks and grade calculation
- Handling user input
- Displaying the final result

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**

Test Case	Expected Result
Enter Student details	Student information stored correctly
Enter Student marks	Marks accepted and saved properly
Calculate grade	Correct total,percentage,and grade displayed
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the system produces correct results for student marks and grade calculation. The application helps in managing student academic records efficiently.

All the stages of the project such as problem analysis, system design, program development, and testing are properly documented. The documentation explains the working process of the system and helps users understand the project clearly.

4.6. Algorithm

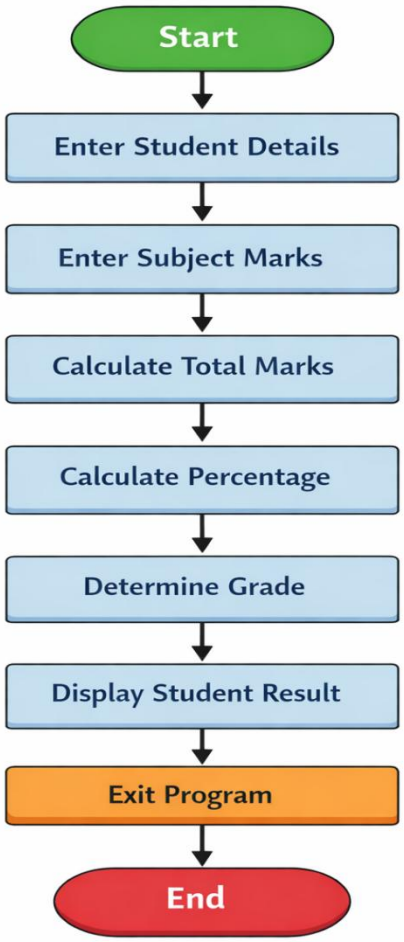
Algorithm for the Student Academic Performance Management System with Grade Calculation:

1. Start the program.
2. Enter student details (name, register number, etc.).
3. Enter the marks for each subject.
4. Store the marks entered by the user.
5. Calculate the total marks of all subjects.
6. Calculate the percentage based on total marks.
7. Determine the grade based on the percentage.
8. Display the student details, total marks, percentage, and grade.
9. End the program.

4.7. Flowchart

Students should draw a flowchart showing the program flow of the Student Academic Performance Management System with Grade Calculation.

Example flow:



5. SOURCE CODE

```
import java.util.Scanner;
 public class StudentPerformanceSystem {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String name;
        String regNo;

        int tamil, english, dm, javaTheory, javaLab, idt;
        int total;
        double percentage;
        char grade;

        System.out.println("----- Student Academic Performance Management System -----");

        // Student details
        System.out.print("Enter Student Name: ");
        name = sc.nextLine();

        System.out.print("Enter Register Number: ");
        regNo = sc.nextLine();

        // Marks entry
        System.out.print("Enter Tamil Marks: ");
```

```
tamil = sc.nextInt();

System.out.print("Enter English Marks: ");
english = sc.nextInt();

System.out.print("Enter Discrete Mathematics Marks: ");
dm = sc.nextInt();

System.out.print("Enter Java Theory Marks: ");
javaTheory = sc.nextInt();

System.out.print("Enter Java Programming Lab Marks: ");
javaLab = sc.nextInt();

System.out.print("Enter Innovation & Design Thinking Marks: ");
idt = sc.nextInt();

// Total calculation
total = tamil + english + dm + javaTheory + javaLab + idt;

// Percentage
percentage = total / 6.0;

// Grade calculation
if (percentage >= 90)
    grade = 'A';
else if (percentage >= 75)
    grade = 'B';
else if (percentage >= 60)
```

```
            grade = 'C';
        else if (percentage >= 50)
            grade = 'D';
        else
            grade = 'F';

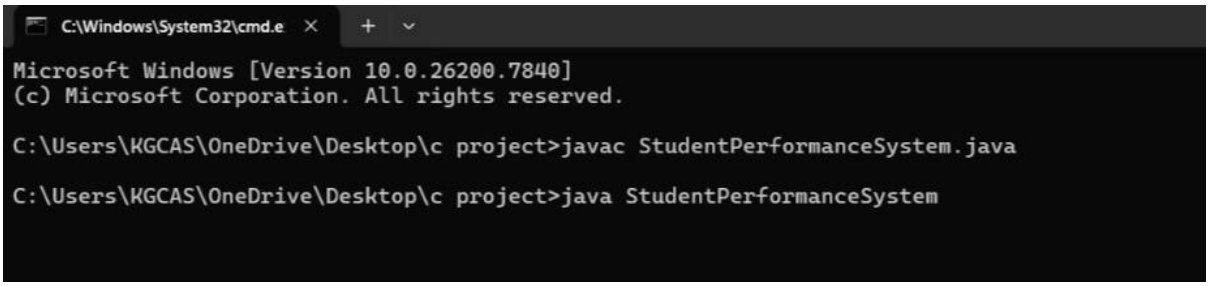
        // Display result
        System.out.println("\n------ STUDENT RESULT ------");
        System.out.println("Name: " + name);
        System.out.println("Register Number: " + regNo);

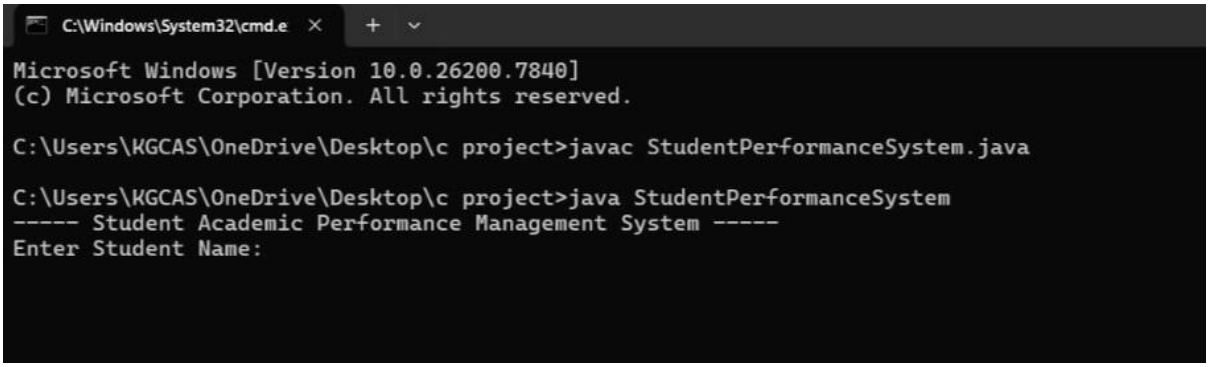
        System.out.println("\nSubject Marks:");
        System.out.println("Tamil: " + tamil);
        System.out.println("English: " + english);
        System.out.println("Discrete Mathematics: " + dm);
        System.out.println("Java Theory: " + javaTheory);
        System.out.println("Java Programming Lab: " + javaLab);
        System.out.println("Innovation & Design Thinking: " + idt);

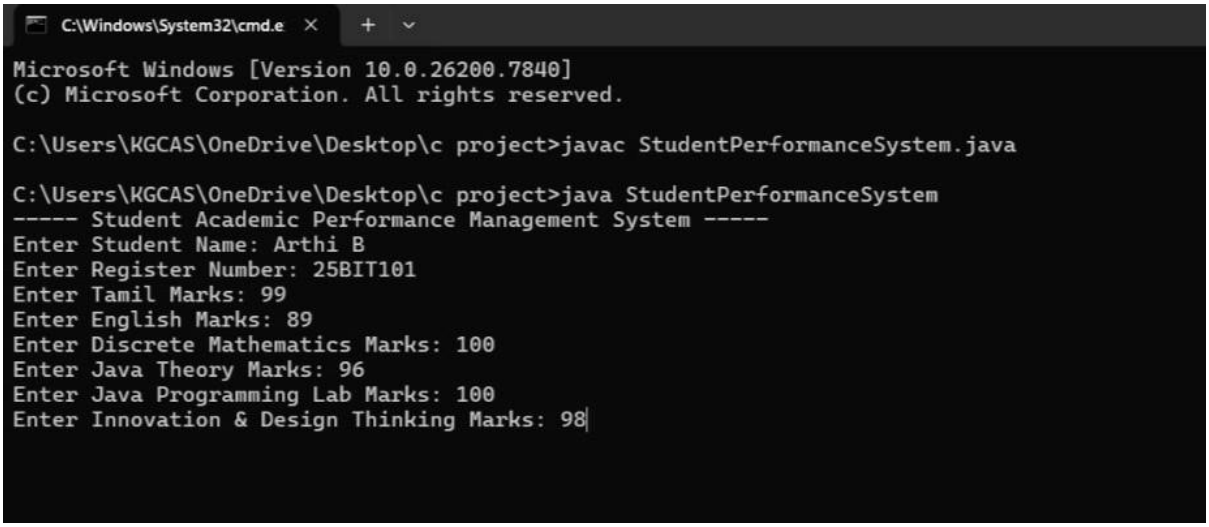
        System.out.println("\nTotal Marks: " + total);
        System.out.println("Percentage: " + percentage + "%");
        System.out.println("Grade: " + grade);

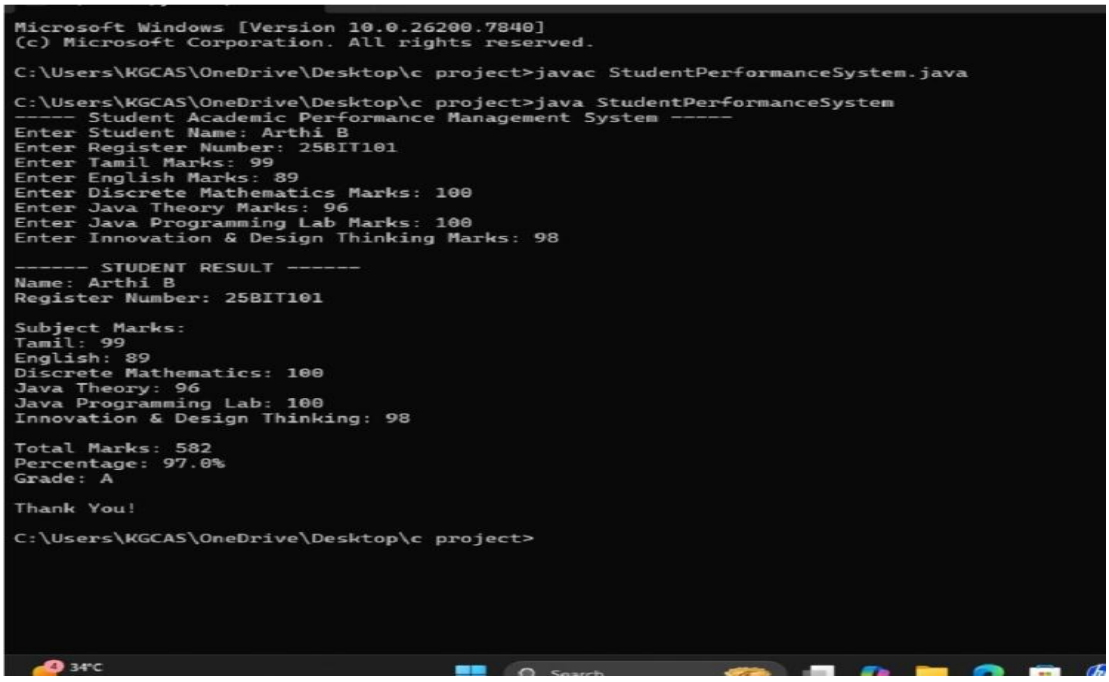
        System.out.println("\nThank You!");
    }
}
```

5. INPUT AND OUTPUT SCREEN






C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.
C:\Users\KGCAS\OneDrive\Desktop\c project>javac StudentPerformanceSystem.java
C:\Users\KGCAS\OneDrive\Desktop\c project>java StudentPerformanceSystem
----- Student Academic Performance Management System -----
Enter Student Name: Arthi B
Enter Register Number: 25BIT101
Enter Tamil Marks: 99
Enter English Marks: 89
Enter Discrete Mathematics Marks: 100
Enter Java Theory Marks: 96
Enter Java Programming Lab Marks: 100
Enter Innovation & Design Thinking Marks: 98


Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.
C:\Users\KGCAS\OneDrive\Desktop\c project>javac StudentPerformanceSystem.java
C:\Users\KGCAS\OneDrive\Desktop\c project>java StudentPerformanceSystem
----- Student Academic Performance Management System -----
Enter Student Name: Arthi B
Enter Register Number: 25BIT101
Enter Tamil Marks: 99
Enter English Marks: 89
Enter Discrete Mathematics Marks: 100
Enter Java Theory Marks: 96
Enter Java Programming Lab Marks: 100
Enter Innovation & Design Thinking Marks: 98
------ STUDENT RESULT ------
Name: Arthi B
Register Number: 25BIT101
Subject Marks:
Tamil: 99
English: 89
Discrete Mathematics: 100
Java Theory: 96
Java Programming Lab: 100
Innovation & Design Thinking: 98
Total Marks: 582
Percentage: 97.0%
Grade: A
Thank You!
C:\Users\KGCAS\OneDrive\Desktop\c project>
34°C

6. DISCUSSION AND CONCLUSION

7.1 Discussion

The Student Academic Performance Management System with Grade Calculation was developed to simplify the process of managing student academic records. The system allows the user to enter student details and subject marks, and it automatically calculates the total marks, percentage, and grade.

During the development of this project, core Java programming concepts such as loops, conditional statements, variables, and input handling using Scanner were used. These concepts helped in performing calculations and displaying the student results correctly.

The system was tested with different marks to ensure that the calculations were accurate. The results show that the system works properly and reduces manual effort in calculating student performance.

7.2 Conclusion

The Student Academic Performance Management System with Grade Calculation successfully demonstrates the use of Core Java programming concepts in developing a simple academic management application.

The system helps to calculate total marks, percentage, and grade automatically, which reduces calculation errors and saves time. It also helps in maintaining student academic information in a simple and organized manner.

This project helps students understand how Java programming can be used to develop practical academic applications. In the future, the system can be improved by adding features such as database storage, graphical user interface (GUI), and report generations.

8. REFERENCE

The following resources were used during the development of the **Student Academic Performance Management System with Grade Calculation** project.

Books

Java: **The Complete Reference – Herbert Schildt**

Websites:

1. **Oracle Java Documentation – https://www.oracle.com/java**
2. **GeeksforGeeks** – **https://www.geeksforgeeks.org**
3. **W3Schools Java Tutorial – https://www.w3schools.com**
4. **Javatpoint Java Tutorial – https://www.javatpoint.com**